

K. Aditya Bharadwaj Nemana

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Education

PhD	McGill University , Mechanical Engineering GPA: 3.9/4.0 • Relevant Courses: Turbulence, Numerical Analysis I, and Matrix Computations. • Thesis: Matrix-free preconditioners for Jacobian-free Newton-Krylov methods applied to turbulent flows. • Awards: Recipient of the McGill Engineering Doctoral Award (MEDA).	Montreal, Canada Sept 2021 – present
MSc	McGill University , Mechanical Engineering GPA: 4.0/4.0 • Relevant Courses: Computational Aerodynamics, Computational Gas Dynamics, Turbomachinery and Propulsion, Dynamics of Combustion, and Finite Element Methods in CFD. • Thesis: Perturbation Signal Tracking Techniques for Unsteady Shock Wave Flows. • Awards: Recipient of the M. P. Paidoussis Prize for the best MSc thesis in the Department of Mechanical Engineering at McGill University in 2022.	Montreal, Canada Sept 2019 – Feb 2022
B.Tech	National Institute of Technology Kurukshetra , Mechanical Engineering GPA: 9.45/10.0	Kurukshetra, India July 2014 – May 2018

Academic Experience

McGill University , Doctoral Candidate • Improved robustness of the Newton-Krylov solver in Bombardier's unstructured CFD code, DRAGON, by introducing backtracking line search. • Introduced a Jacobian-free Newton-Krylov framework with matrix-free preconditioners in DRAGON for applications in RANS simulations. • Developed a matrix-free preconditioner based on the Newton-Schulz iterative method.	Montreal, Canada Sept 2021 – present
McGill University , Teaching Assistant • Conducted tutorials, graded assignments, and provided feedback to students. • Assisted with the development of course materials and resources.	Montreal, Canada Sept 2021 – present
Indian Institute of Technology (IIT) Kanpur , Research Associate • Worked on the design and development of a miniature power-generation system using liquid fuel. • Conducted computational fluid dynamics (CFD) analyses using ANSYS Fluent and ICEM CFD to evaluate system performance, efficiency, and optimize the heat-sink design.	Kanpur, India July 2018 – May 2019

Professional Experience

Post-Graduate Students' Society (PGSS), McGill University , Equity and Diversity Commissioner Accessibility Commissioner • Actively serving on McGill University's EDI Committee (EDIC) and co-chairing the PGSS Equity Committee.. • Introduced EDI Awards aimed at supporting projects and initiatives that promote equity, diversity, and inclusion within the graduate student community. • Developed and implemented an accessibility survey for graduate students and postdoctoral researchers at McGill University.	Montreal, Canada May 2024 – present
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Office of E-IDEA, McGill University, SCOPE Fellow

- Member of the Social Impact, Community Building, and Outreach Partnerships in Engineering (SCOPE) team.
- Designed and developed modules to help graduate students integrate EDI principles into engineering design.

Montreal, Canada
Jan 2025 – present

Office of E-IDEA, McGill University, Summer Intern

- Researched and compiled approaches to integrating EDI principles into engineering design.
- Created and maintained a database of underrepresented engineers from marginalized communities and engineering innovations by indigenous peoples of Canada.

Montreal, Canada
May 2024 – Aug 2024

Volunteering Experience

Subcommittee for Persons with Disabilities, McGill University, Co-Chair

- Organized regular committee meetings to discuss and address the needs of persons with disabilities.
- Represented the committee in meetings with university administration and other stakeholders to advocate for the rights and needs of persons with disabilities.

Montreal, Canada
Sept 2025 – present

Graduate Engineering Equity Committee (GEEC), McGill University, Events Coordinator

- Organized monthly EDI-focused events, including movie nights, guest speaker talks, and panel discussions.
- Managed event logistics, including registration forms, posters, room bookings, advertising, and feedback collection.

Montreal, Canada
Mar 2022 – present

Council of Graduate and Postdoctoral Studies (CGPS), McGill University, Engineering Graduate Student Representative

- Contributed to the development, revision, and approval of the regulations, guidelines, and policies on graduate and postdoctoral studies.
- Participated in strategic planning for initiatives to promote and enhance graduate and postdoctoral studies.

Montreal, Canada
Sept 2023 – Apr 2025

Skills

Programming and Tools: Python, C/C++, Git, MATLAB, Bash, Markdown, and Microsoft Office.

Computational Tools: ANSYS Fluent, ICEM CFD, CATIA, FreeCAD.

Research Skills: Computational Fluid Dynamics (CFD), numerical methods, and applied mathematics.

Libraries and Frameworks: PETSc, Eigen, NumPy, SciPy, Matplotlib.

Languages: English, French (Beginner), Hindi, Telugu, and Odia.

Publications

On the disturbance tracking methods for shock wave reflection studies

Sept 2025

Nemana, Kameswara Aditya Bharadwaj, Timofeev, Evgeny

[10.1063/5.0286049](#) (AIP Conference Proceedings)

Development of a Standalone, Liquid-Fueled Miniature Power-Generation System

Sept 2019

Bharadwaj, N. K. A., Jain, N., Arghode, V. K.

[10.1115/1.4044764](#) (Journal of Energy Resources and Technology)